# ViPragSent-Final: A Development-Selected Ensemble for Vietnamese Pragmatic Sentiment Evaluation

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## Abstract

Pragmatic interpretation in Vietnamese-language text can depend on signals beyond surface polarity. We evaluate six pragmatic phenomena in the fixed ViPragSent setting, which has a documented social-media source component and a private-data boundary. ViPragSent-Final is a frozen label-wise ensemble selected with five-fold development out-of-fold evidence: it combines retained ViSoBERT and PhoBERT sources, probability blends for irony and code-switching, and a targeted mocking component. On one post-freeze canonical-test evaluation, it reaches 84.39 macro pragmatic F1, 1.56 points above Vistral-7B, the strongest complete best-tuned baseline. The aggregate lead does not establish strict per-label dominance: the final system remains 0.40 points below the strongest irony result, 0.11 below the strongest idiom/figurative-language result, and 0.13 below the strongest code-switching result. Stored predictions, configuration, and artifact hashes are traceable, but this is not an independent training reproduction; raw source text remains private.

## 1 Introduction

Vietnamese social-media language is a consequential setting for affective NLP: users express evaluations in short, informal, context-dependent text, while platforms and downstream analyses often reduce those expressions to surface polarity. That reduction is useful for ordinary sentiment tasks, but it is insufficient when the communicative force of an utterance depends on implication, incongruity, figurative language, language alternation, or ridicule. Vietnamese pretrained and social-media-oriented models provide valuable modelling context (Nguyen and Tuan Nguyen, 2020; Conneau et al., 2020; Nguyen et al., 2023b), and Vietnamese sentiment and emotion resources have enabled important evaluations (Nguyen et al., 2018; Ho et al., 2020; Tran et al., 2026). However, their task definitions do not jointly evaluate the pragmatic phenomena considered here.

Pragmatic sentiment is not a single extension of polarity. Implicit sentiment can convey evaluation without an overt sentiment word; sarcasm and irony can reverse or complicate literal cues; idioms and other figurative expressions can require culturally grounded interpretation; code-switching can alter the lexical and discourse signals available to a classifier; and mocking can signal stance through ridicule rather than an explicit polarity marker. Work on verbal irony and Vietnamese figurative-language evaluation reinforces the importance of context and phenomenon-specific coverage (Ghosh et al., 2019; ReML-AI, 2026). These phenomena can co-occur and remain difficult to capture with a generic sentiment benchmark. At the same time, this observation is not a claim that prior Vietnamese resources are inadequate for their intended purposes, nor that the six labels exhaust Vietnamese pragmatics.

The empirical gap is therefore specific: a provenance-bounded evaluation is needed that brings these six phenomena into one fixed Vietnamese setting, distinguishes them from auxiliary polarity and emotion labels, and compares several realistic model families under the same protocol. ViPragSent supplies 12,000 adjudicated records, with an 8,000/2,000/2,000 train/development/test split. Its gold corpus comprises 10,000 retained local ViSoBERT-export records and 2,000 VIVID-derived, context-augmented candidate records; source text remains private and the latter provenance retains authorization and licence limitations. The benchmark is consequently an evaluation resource with explicit access boundaries, not a public claim to redistribute all underlying text.

The comparison also needs a distinction between model family and evaluation result. Auxiliary-task learning can jointly model related linguistic signals with targeted sentiment, but its benefit is an empirical question rather than a default expectation (Moore and Barnes, 2021). Likewise, prompted and instruction-tuned models require task- and language-specific evaluation rather than assumed superiority (Toukmaji and Flanigan, 2025; Dou et al., 2024). We therefore retain complete best-tuned baselines from PhoBERT, XLM-R-large, Sailor-7B, and Vistral-7B, and assess a frozen label-wise ViPragSent-Final ensemble. The final system is not offered as a novel ensemble architecture: it is a development-selected composition designed to test whether different retained sources and thresholds yield a stronger aggregate result while keeping label-level trade-offs visible.

The selection boundary is central. ViPragSent-Final uses five-fold development out-of-fold evidence to select its label-wise sources and thresholds, freezes the configuration, and then evaluates it once on the canonical test partition. This keeps development selection distinct from test evaluation, while the stored predictions, configuration, and artifact hashes support artifact-level traceability rather than an independent training rerun. Historical 73.7469 ViPragSent configurations, ablations, and diagnostics remain useful versioned evidence, but they are not interchangeable with the promoted final system. Similarly, the stopped true-anchor arbiter cycle did not access canonical test labels and is not promoted.

This study addresses two questions:

**RQ1.** How does ViPragSent-Final compare overall and label by label with the complete retained best-tuned baselines on the fixed pragmatic evaluation?

**RQ2.** What does the final system's mixed pattern of label-wise leadership reveal about interpreting an aggregate macro-F1 gain?

The answers require both an aggregate comparison and transparent exceptions. ViPragSent-Final reaches 84.3883 macro pragmatic F1 on the post-freeze canonical test evaluation. Vistral-7B, the strongest complete best-tuned baseline, reaches 82.8250, yielding an aggregate difference of +1.5633 points. This result does not establish universal superiority: the final system remains below the strongest retained baseline result for irony, idiom/figurative language, and code-switching. RQ2 therefore treats aggregate performance as a summary to be interpreted alongside label-specific leadership, not as a substitute for it.

The paper makes five contributions:

1. It defines a six-label Vietnamese pragmatic-sentiment evaluation with separate polarity and emotion auxiliaries, fixed splits, and a private-data governance boundary.
2. It documents the annotation, adjudication, provenance, and access limitations needed to interpret the 12,000-record gold corpus.
3. It provides a common comparative evaluation across Vietnamese, multilingual, social-media-oriented, 7B instruction-tuned, and prompted baseline families, with explicit seed/run qualifications.
4. It reports ViPragSent-Final as a frozen, development-selected label-wise system evaluated once after freeze, rather than as an unqualified architectural advance.
5. It reports both the +1.5633 aggregate advantage and all label-level gaps, linking the resulting claims to predictions, configuration, and validation artifacts.

Section 2 positions these choices against related resources and model families. Sections 3–5 specify the task, protocol, and comparison; the analysis then separates final-system conclusions from historical and exploratory diagnostics.

## 2 Related Work

ViPragSent-Final sits between Vietnamese language modelling, affective-resource construction, and pragmatic-language evaluation. Prior work supplies model families, datasets, and task formulations, but none of the cited work by itself establishes the present six-label evaluation, the final ensemble's aggregate result, or a general advantage for a particular training strategy. We therefore separate related tasks from this paper's fixed evaluation and provenance-bounded contribution.

### 2.1 Vietnamese Pretraining and Social-media Language

PhoBERT provides a Vietnamese monolingual pretrained encoder, while XLM-R provides a multilingual representation-learning reference (Nguyen and Tuan Nguyen, 2020; Conneau et al., 2020). ViSoBERT extends this landscape with a Vietnamese social-media language-model context (Nguyen et al., 2023b). These studies motivate comparing monolingual, multilingual, and social-media-oriented encoders, but their pretraining objectives and benchmarks do not define the present pragmatic labels. Accordingly, ViPragSent evaluates these families under one fixed protocol rather than treating social-media pretraining as a guarantee of label-wise superiority.

### 2.2 Vietnamese Affective Resources and Pragmatic Phenomena

Vietnamese affective resources have principally targeted ordinary sentiment or emotion. UIT-VSFC is a student-feedback sentiment corpus, and UIT-VSMEC is a social-media emotion resource (Nguyen et al., 2018; Ho et al., 2020); ViGoEmotions further demonstrates fine-grained emotion classification on Vietnamese social-media comments (Tran et al., 2026). These resources establish useful task and domain precedents, but their label inventories do not jointly operationalize implicit sentiment, sarcasm, irony, figurative language, code-switching, and mocking. In this study they remain contextual or external-diagnostic resources, not sources of ViPragSent records.

Pragmatic interpretation is not reducible to ordinary polarity: verbal irony can depend on semantic incongruity and contextual strategies (Ghosh et al., 2019), while figurative-language evaluation requires its own linguistic and cultural coverage. The VIVID benchmark explicitly targets Vietnamese figurative-language gaps (ReML-AI, 2026). These lines of work motivate treating pragmatic labels as separate evaluation targets; they do not support a claim that the present corpus exhausts Vietnamese pragmatics. ViPragSent instead fixes a bounded six-label task and reports the provenance of its VIVID-derived candidate stratum separately.

### 2.3 Social-media Variation and Code-switching

Social-media-oriented models and emotion resources show that informal Vietnamese introduces domain-specific variation beyond conventional review or feedback data (Nguyen et al., 2023b; Ho et al., 2020; Tran et al., 2026). Code-switching adds a further phenomenon whose evaluation cannot be inferred from ordinary sentiment scores alone. The present work therefore includes code-switching as one independently scored pragmatic label, rather than using a generic social-media claim as a substitute for evidence. This design leaves broader questions about cross-domain generalization and code-switching mechanisms open.

### 2.4 Auxiliary Objectives, Ensembles, and Leakage-safe Selection

Multi-task work has shown that related linguistic objectives can be jointly modelled with targeted sentiment classification (Moore and Barnes, 2021). That precedent motivates evaluating auxiliary objectives, but it does not imply that a fixed auxiliary set benefits a new corpus or backbone. ViPragSent therefore preserves its historical ablations as configuration-specific evidence and does not attribute the promoted result to any single auxiliary task.

The promoted system is not presented as a novel ensemble architecture. It is a frozen label-wise composition selected with five-fold development out-of-fold evidence and evaluated once after freeze on canonical test labels. This protocol distinguishes development selection from test evaluation; it is not a claim that no prior work uses specialization or ensembling. Its contribution is the documented, leakage-bounded application to the fixed ViPragSent task, including transparent aggregate and per-label comparisons.

### 2.5 Instruction-tuned Baselines in Low-resource Evaluation

Instruction-tuned and prompted models provide useful comparison families, but their behaviour in lower-resource settings is empirical rather than assumed. A controlled cross-lingual study compares prompting, translation, fine-tuning, and instruction tuning across target languages and downstream tasks (Toukmaji and Flanigan, 2025); Sailor supplies an open South-East Asian language-model family (Dou et al., 2024), while Vistral supplies the Vietnamese instruction-tuned model card used for the configured baseline (Nguyen et al., 2023a). These sources motivate reporting prompted and 7B instruction-tuned baselines with their recorded scope. They do not establish that an LLM family must win Vietnamese pragmatic detection, nor that a repository-specific QLoRA run is independently reproducible.

For AIVIVN, Nguyen et al. provide a proceedings description of a binary Vietnamese e-commerce-review setting (Nguyen et al., 2020).

We cite it only for that published setting: the repository's evaluated artifact is a separately hash-identified Kaggle mirror used for external polarity diagnostics, not an asserted canonical organizer release. Across these strands, the gap addressed here is a provenance-bounded comparative evaluation of six Vietnamese pragmatic phenomena, with a development-selected final system whose aggregate lead is explicitly separated from per-label leadership.

## 3 Task and Data

### 3.1 Task Formulation and Labels

ViPragSent treats pragmatic-sentiment analysis as a structured prediction problem over a Vietnamese-language record in the documented source composition. Each record carries a multi-label pragmatic vector, an intended-polarity label, and an emotion label. The pragmatic vector is evaluated label by label, so phenomena may co-occur rather than forming mutually exclusive classes. This formulation deliberately separates how sentiment is pragmatically conveyed from the intended valence and affective category assigned to the same record.

The six binary pragmatic labels are *implicit sentiment*, *sarcasm*, *irony*, *idiom/figurative language*, *code-switching*, and *mocking*. Current project documentation records the author-team-defined operational criteria: implicit sentiment is indirectly conveyed evaluation or emotion; sarcasm is a surface stance opposite to intended stance; and irony is an incongruity between expectation and occurrence that need not reverse polarity. It records idiom/figurative language as a fixed idiom or non-literal expression, code-switching as the presence of at least one English content word rather than a common borrowed term or abbreviation, and mocking as ridicule or demeaning treatment of a person, group, or action. Each field is stored independently and can co-occur with the others; an affirmative value records the presence of its named operational target rather than selection of one mutually exclusive pragmatic category. The criteria guide contextual, interpretive judgment rather than a deterministic cue list. They are a project-specific operational taxonomy, not a claim that these six fields exhaust Vietnamese pragmatics or instantiate a standard literature taxonomy.

Intended polarity is a separate three-way auxiliary label with positive, neutral, and negative values. It is assigned from intended meaning rather than surface praise, which is important when ironic or sarcastic wording reverses literal valence. The emotion auxiliary is likewise distinct from the pragmatic vector and uses the repository's UIT-VSMEC-style label set: enjoyment, sadness, anger, disgust, fear, surprise, and other. Pragmatic labels therefore identify communicative phenomena, while polarity and emotion characterize complementary ordinary sentiment and affect targets; neither auxiliary is a pragmatic label or a proxy for one.

Table 1 records the verified task and corpus inventory used in this study. Appendix B reports positive pragmatic-label counts by fixed split; because the labels are multi-label, these counts are not mutually exclusive class totals.

### 3.2 Data Provenance, Splits, and Governance

The gold corpus contains 12,000 adjudicated records. Its stored source metadata identifies 10,000 retained local ViSoBERT-export records and 2,000 author-created, context-augmented derivatives based on Vietnamese idiom/proverb materials from VIVID (ReML-AI, 2026). The contextualization was informed in part by verbal-irony strategies described by Ghosh et al. (2019); it is not presented as a reproduction of that English-language study. Those 2,000 rows replace earlier ViSoBERT rows and are present in the fixed gold partitions used for all reported experiments; the replaced rows are not corpus members. Its fixed split is 8,000 training records, 2,000 development records, and 2,000 test records. The source-composition report records 6,667/1,333 local/generated-candidate records in train, 1,667/333 in development, and 1,666/334 in test, respectively; these are source metadata strata, not mutually exclusive pragmatic classes. The build report records no missing reviews, unresolved disagreements, or invalid records in the final gold build. The repository's fixed partitions determine all reported experiments; this paper neither infers a sampling protocol nor alters membership. Table 1 traces corpus size and split fields to the gold-build report, rather than to the earlier silver-label registry.

The source-text boundary is restrictive. Raw and processed source text remains private and is not redistributed; this manuscript neither claims a public release nor reproduces source-text examples. Source authorization and license documenta-

| Component | Verified labels or value | Role |
|---|---|---|
| Pragmatic vector | implicit_sentiment; sarcasm; irony; idiom_figurative; code_switching; mocking | Binary, multi-label target |
| Intended polarity | positive; neutral; negative | Auxiliary target |
| Emotion | enjoyment; sadness; anger; disgust; fear; surprise; other | Auxiliary target |
| Gold corpus | 12,000 adjudicated records | ViPragSent evaluation corpus |
| Gold-text composition | 10,000 retained local social-media export rows; 2,000 author-created, context-augmented derivatives based on VIVID idiom/proverb materials | Fixed experimental corpus |
| Fixed split | 8,000 / 2,000 / 2,000 (train/dev/test) | Experimental partitions |
| Access boundary | Private, non-commercial research only; no public raw-text release | Data governance |

Table 1: Verified ViPragSent task and data inventory.

tion are pending project action. The release boundary applies to text access, not to the experimental artifacts and provenance records used for the reported evaluation.

The repository also records UIT-VSFC, UIT-VSMEC, and an AIVIVN-2019 Kaggle mirror as external evaluation resources. They are not incorporated into ViPragSent. Their role is confined to the external ordinary-task diagnostic; the manuscript reports aggregate results without distributing their text or treating a mirror as a canonical release. Their exact-artifact source authorization remains pending in the project governance record.

### 3.3 Annotation and Agreement

The final labels are post-adjudication labels. Two reviewers and a fixed adjudicator are listed in the agreement artifact, and the adjudicator's labels define the experiment gold split. Before adjudication, the two reviewers differed on at least one of the six pragmatic, polarity, or emotion fields for 5,261 of the 12,000 shared records; the fixed adjudicator resolved these cases when selecting the final gold labels. The reported agreement values therefore summarize a post-adjudication workflow; they must not be interpreted as an independent, blinded third-rater study. This distinction also means that recomputing an agreement statistic cannot be used to replace or refresh the fixed experimental gold labels.

Across the recorded label families, the macro mean pairwise agreement is 0.93, with macro mean pairwise Cohen's $\kappa$ of 0.82, Fleiss' $\kappa$ of 0.82, and Krippendorff's $\alpha$ of 0.82. These values provide a compact description of the recorded annotation artifact, not an estimate of agreement from a newly collected annotation study. Table 4 in Appendix A provides the detailed per-label values, keeping the main section focused on the task, provenance, and governance conditions.

## 4 Systems and Evaluation

### 4.1 Baseline System Families and Original ViPragSent

The complete retained baselines are PhoBERT, XLM-R-large, Sailor-7B, and Vistral-7B. PhoBERT and XLM-R-large are encoder fine-tunes; the encoder configuration uses learning rate $2 \times 10^{-5}$, maximum sequence length 128, effective batch size 32, up to 10 epochs, and patience-two early stopping. Sailor-7B and Vistral-7B are NF4 four-bit QLoRA SFT systems (learning rate $10^{-4}$, length 256, effective batch size 16, three epochs, LoRA $r = 16$, $\alpha = 32$, dropout 0.05). Their base revisions are recorded as download-time provenance rather than independently logged runtime fields. Learned encoder and 7B results retain their recorded three-seed scope; prompted API baselines are single recorded runs. These differences in scale and tuning budget are reported rather than treated as equal compute.

Original and previous optimized ViPragSent configurations, the archived incumbent (84.0904), and historical ablations are progression evidence only. They are not independent external baselines and are not substituted for the promoted system.

### 4.2 ViPragSent-Final Label-wise Ensemble

ViPragSent-Final is a frozen label-wise prediction ensemble, not a single encoder. The configured retained ViSoBERT sources `visobert_20260902` and `visobert_20260903` supply implicit sentiment and sarcasm, respectively; irony and code-switching use probability $0.6\,p_{\text{ViSoBERT}} + 0.4\,p_{\text{PhoBERT}}$; uniform three-seed ViSoBERT supplies idiom/figurative language; and a targeted uniform three-seed component supplies mocking. This description makes the system auditable and does not attribute its performance to an individual component or claim architectural novelty.

### 4.3 Development-only Selection, OOF Procedure, and Freeze

Candidates were generated and compared on development data only through deterministic five-fold multilabel OOF evaluation. The stored objective is mean target-label F1 minus 0.25 times cross-fold standard deviation; test labels are explicitly recorded as unused for selection. Rejected development candidates include character TF–IDF code-switch experts, a residual specialist, rank blending, and fixed-weight/logistic stackers. After source selection, thresholds were refit on full development data and frozen at 0.55, 0.87, 0.96, 0.99, 0.50, and 0.35 for implicit sentiment, sarcasm, irony, idiom/figurative language, code-switching, and mocking.

The frozen configuration was evaluated once on the canonical 2,000-record test partition. Saved prediction IDs exactly match canonical test IDs. Promotion required a strictly higher macro score than the incumbent, an improved target, and no label drop exceeding 0.10 points.

### 4.4 Metrics and Statistical Protocol

For each pragmatic label, binary macro-F1 averages negative- and positive-class F1. Macro pragmatic F1 is the equal-weight mean over the six labels, so aggregate leadership does not imply per-label leadership. Repository evaluation uses 1,000 bootstrap resamples for 95% intervals; stored paired comparisons use a paired nonparametric bootstrap over aligned test records with 1,000 resamples. Historical intervals and tests remain scoped to their named systems and comparisons. Three-seed normal intervals are descriptive, and no unrecorded multiple-comparison correction is claimed for the development label-wise search.

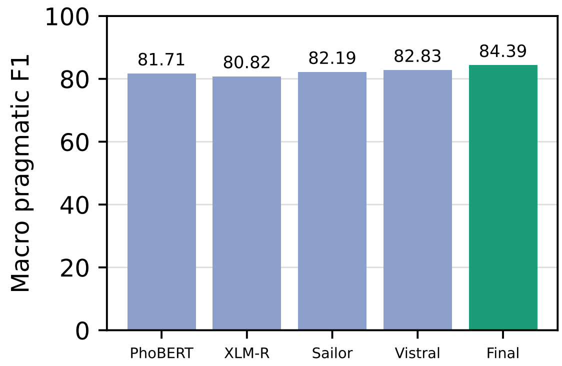


Figure 1: Macro pragmatic F1 for the complete best-tuned baseline systems and ViPragSent-Final. The comparison is an aggregate, equal-weight metric over six pragmatic labels.

### 4.5 Reproducibility, Compute, and Artifact Traceability

The promoted bundle records source assignments, threshold metadata, prediction files, an archived incumbent, and artifact hashes. Its 53 indexed files match after CRLF-to-LF transport normalization. Nine checkpoint paths and hashes are recorded but checkpoint files are unavailable locally. The result is therefore validated at artifact level, not independently reproduced by retraining. Legacy ablation, source-stratified, low-resource, calibration, and external diagnostics retain their original system-version scope and are not final-ensemble evidence unless explicitly re-evaluated.

## 5 Results

### 5.1 Final Aggregate Comparison

Table 2 reports the complete best-tuned baseline comparison and ViPragSent-Final. ViPragSent-Final reaches 84.3883 macro pragmatic F1, compared with 82.8250 for Vistral-7B, the strongest complete baseline. The resulting aggregate difference is +1.5633 points. Within the evaluated systems and this fixed evaluation, this is the highest recorded aggregate macro pragmatic F1.

### 5.2 Label-wise Comparison

The aggregate comparison and the per-label-leader comparison answer different questions and are kept separate. Relative to the strongest best-tuned baseline result for each label, ViPragSent-Final is ahead by 7.1359 points on implicit sentiment, 0.2463 on sarcasm, and 0.0726 on mocking. It is below the label leader by 0.3955 points on irony,

| System | Implicit | Sarcasm | Irony | Idiom/fig. | Code-switch. | Mocking | Macro |
|---|---|---|---|---|---|---|---|
| PhoBERT best-tuned | 60.85 | 77.37 | 96.32 | **97.30** | 78.99 | 79.45 | 81.71 |
| XLM-R-large best-tuned | 58.30 | 79.49 | 96.85 | **97.30** | 72.16 | 80.80 | 80.82 |
| Sailor-7B best-tuned | 57.43 | 78.57 | **97.41** | 97.15 | 81.28 | 81.31 | 82.19 |
| Vistral-7B best-tuned | 59.59 | 80.03 | 97.05 | 96.35 | **81.95** | 81.98 | 82.83 |
| ViPragSent-Final | **67.98** | **80.28** | 97.02 | 97.18 | 81.82 | **82.05** | **84.39** |

Table 2: Final pragmatic results (binary macro-F1 percentages). Bold indicates the best value in each column among the compared best-tuned baseline systems and ViPragSent-Final. The final system is a frozen, development-selected label-wise ensemble evaluated once on the canonical test set; the aggregate result does not imply per-label dominance.

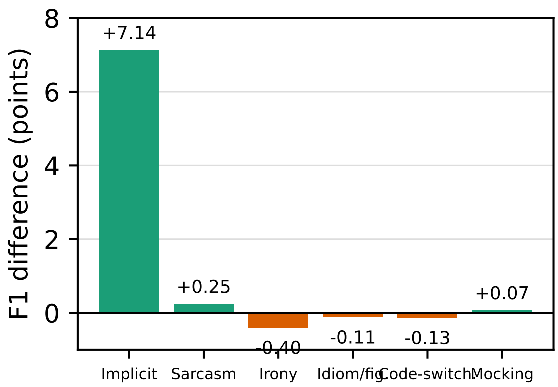


Figure 2: Per-label difference between ViPragSent-Final and the strongest best-tuned baseline result for each label. The zero line distinguishes positive margins from localized deficits; the vertical scale is not truncated to exaggerate small differences.

0.1138 on idiom/figurative language, and 0.1294 on code-switching (Table 3).

### 5.3 Uncertainty and Paired Comparisons

The promoted bundle contains a post-hoc paired bootstrap over fixed test predictions for the final continuation versus its archived incumbent, using 1,000 replicates. The observed macro difference is +0.2979 points with a 95% bootstrap interval of [-0.0714, 0.6463]; 94.4% of replicates are positive. This interval is descriptive and was not used for selection. Comparable aligned prediction files for a paired final-versus-Vistral test and final-versus-each-label-leader tests are not present in the promoted bundle. We therefore report their point-estimate differences without a significance claim or invented interval, and do not apply Holm correction because no formal multi-hypothesis family is reported.

### 5.4 Interpretation of Mixed Leadership

The result is an aggregate-performance lead, not strict dominance of every pragmatic label. All three localized deficits are below 0.40 points, while the implicit-sentiment gain is 7.1359 points and sarcasm and mocking also have positive margins. Thus, the aggregate advantage is driven primarily by the large implicit-sentiment gain, supplemented by smaller gains on sarcasm and mocking. These gains coexist with three localized deficits, producing the highest equal-weight macro pragmatic F1 among the evaluated complete systems.

### 5.5 Development Progression and Legacy Analyses

The progression from ViPragSent original (73.7469) through the previous optimized system (82.0699) and final incumbent (84.0904) to the promoted continuation (84.3883) is shown in Appendix E. These rows document development history and are not additional external baselines.

Existing ablation, source-stratified, low-resource, external-diagnostic, calibration, confusion, and annotation-agreement artifacts remain valid only for the system version named in each artifact. In particular, the older 73.75 ablation behavior must not be attributed to ViPragSent-Final.

### 5.6 Answer to the Research Questions

ViPragSent-Final has the strongest aggregate macro pragmatic F1 among the evaluated systems, with a +1.5633-point margin over Vistral-7B. This bounded aggregate result does not establish strict per-label superiority.

## 6 Analysis

This section separates observations stored in the analysis artifacts from their permitted interpretation. It does not use error, confidence, or low-

| Comparison | Leader / score | ViPragSent-Final | Difference |
|---|---|---|---|
| Aggregate macro F1 | Vistral-7B / 82.8250 | 84.3883 | +1.5633 |
| Implicit sentiment | PhoBERT / 60.8470 | 67.9829 | +7.1359 |
| Sarcasm | Vistral-7B / 80.0318 | 80.2781 | +0.2463 |
| Irony | Sailor-7B / 97.4132 | 97.0177 | -0.3955 |
| Idiom/figurative | PhoBERT/XLM-R-large / 97.2958 | 97.1820 | -0.1138 |
| Code-switching | Vistral-7B / 81.9458 | 81.8164 | -0.1294 |
| Mocking | Vistral-7B / 81.9802 | 82.0528 | +0.0726 |

Table 3: Two comparison definitions. The first row compares aggregate system performance with Vistral-7B, the strongest complete baseline. The remaining rows compare each ViPragSent-Final label score with the strongest best-tuned baseline score for that label.

resource outputs to revise the final-system ordering in Section 5.

### 6.1 Final Aggregate Balance

**OBSERVED EVIDENCE.** ViPragSent-Final has the highest recorded equal-weight macro pragmatic F1 (84.3883) among the compared complete systems, a +1.5633-point aggregate difference from Vistral-7B. Its per-label comparison contains a large implicit-sentiment margin (+7.1359), smaller positive sarcasm and mocking margins, and deficits of -0.3955 on irony, -0.1138 on idiom/figurative language, and -0.1294 on code-switching.

**PLAUSIBLE INTERPRETATION.** The aggregate score can lead despite mixed per-label leadership because it is the equal-weight mean of six binary macro-F1 values. The final label-wise source selection may allow specialized prediction sources to contribute where their development evidence was strongest. This is an interpretation of the stored configuration and values, not a causal isolation of any component.

**UNTESTED HYPOTHESIS.** Alternative source assignments, blend weights, or thresholds could change the balance of margins and deficits. The available evidence does not estimate this sensitivity beyond the stored development-selection procedure.

### 6.2 Final-system Error Profile and Historical Confusion Diagnostic

**OBSERVED EVIDENCE.** Final-system confusion counts are available label by label for the 2,000 canonical test records. The largest false-positive count is implicit sentiment (178), followed by mocking (74) and code-switching (67); the largest false-negative count is sarcasm (154), followed by implicit sentiment (115) and mocking (104). Irony and idiom/figurative language have no recorded false positives and 19 and 18 false negatives, respectively. These counts, together with the recorded label prevalences, show that error behavior differs substantially across labels.

**PLAUSIBLE INTERPRETATION.** The large implicit-sentiment margin can contribute strongly to the equal-weight aggregate gain despite remaining false positives and false negatives. The narrow deficits for irony, idiom/figurative language, and code-switching are consistent with a balance-improving label-wise mixture rather than uniform dominance.

**UNTESTED HYPOTHESIS.** Whether these patterns arise from specialized sources, thresholds, label prevalence, or development-selection complexity would require controlled component ablations evaluated with the frozen final protocol.

**OBSERVED EVIDENCE.** The stored row-normalized pragmatic-polarity confusion matrix is for ViPragSent (ours), using its seed-20260520 prediction file, with six recorded classes: positive, negative, neutral, ironic-positive, sarcastic-negative, and mocking. Rows are gold classes and columns are predicted classes. It is therefore a single-seed diagnostic for this evaluated system and class construction, rather than an error analysis of the other systems or a three-seed aggregate from the main comparison. The source graphic is retained in the verified artifact bundle rather than included in the current appendix, so that its six class labels remain legible when inspected. The recorded neutral diagonal is 79.76%, whereas the negative row assigns 47.06% of its records to neutral; the sarcastic-negative and mocking diagonals are 55.04% and 55.08%, respectively. The ironic-positive row has no counts in this stored matrix, so it cannot support a row-level recall observation. These values are descriptive entries in a row-normalized artifact, not independent estimates for every pragmatic label in the study.

**PLAUSIBLE INTERPRETATION.** The concentration of some negative and pragmatic-polarity records outside their diagonal cells identifies where this particular prediction set can be inspected more closely. It supports a bounded statement that the recorded decision behavior is uneven across the represented classes, without establishing whether the source of an error is wording, annotation boundary, class prevalence, or the learned representation.

**UNTESTED HYPOTHESIS.** A targeted experiment could test whether rebalancing, alternative class construction, or a different prediction interface changes these off-diagonal patterns. No such experiment is present in the artifact bundle, so these possibilities are not used as conclusions about the model architecture or the data.

### 6.3 Calibration

**OBSERVED EVIDENCE.** Calibration is reported only for seed-20260520 prediction files with stored pragmatic-polarity confidence scores: PhoBERT fine-tune, XLM-R-large, and ViPragSent (ours). The artifact excludes systems without those recorded scores, so it does not justify a calibration ranking across all eight main-comparison systems or a three-seed calibration estimate. Each eligible system has 2,000 scored records and zero missing confidence entries; the stored ECE values are 0.09 for PhoBERT fine-tune, 0.07 for ViPragSent (ours), and 0.15 for XLM-R-large. The bin-level reliability outputs and their visual detail are retained in the verified artifact bundle rather than included in the current appendix; they should be read with the corresponding confidence-scope qualification.

**PLAUSIBLE INTERPRETATION.** Within this restricted set of recorded confidence outputs, ViPragSent has the lowest stored ECE. This statement concerns calibration error on the defined pragmatic-polarity target; it neither changes the macro-F1 ordering nor implies that the same ordering would hold for systems without archived confidence values.

**UNTESTED HYPOTHESIS.** Whether calibration post-processing or a different confidence extraction procedure would alter these values is not tested here. Accordingly, no causal account of the ECE differences is made.

### 6.4 Source-stratified Sensitivity

The source-stratified, calibration, confusion, and low-resource analyses below are historical-system diagnostics unless explicitly marked otherwise; they are not analyses of ViPragSent-Final.

**OBSERVED EVIDENCE.** P1 splits the fixed test partition into 1,666 ViSoBERT-local records and 334 VIVID-labelled records, with all reported systems scored separately across three seeds. Appendix Table 6 displays ViPragSent, PhoBERT, and ViSoBERT to make the scale and uncertainty of the source-stratified results inspectable. For ViPragSent, PhoBERT, and ViSoBERT respectively, macro pragmatic F1 is 52.56 [52.38, 52.73], 60.60 [58.63, 62.57], and 62.08 [60.69, 63.47] on the ViSoBERT-local stratum, versus 79.84 [79.71, 79.96], 87.21 [85.48, 88.95], and 93.04 [90.55, 95.54] on the VIVID-labelled stratum. The latter stratum is smaller, so its intervals should be read with its 334-record scope in view.

**PLAUSIBLE INTERPRETATION.** These results describe the recorded, source-labelled composition of this test partition. They do not identify a source-domain cause, establish that one source is intrinsically easier, or generalize to a broader domain-transfer setting.

**UNTESTED HYPOTHESIS.** Differences between strata could reflect the constructed labels, class composition, source provenance, or other coupled properties. The present observational split does not isolate those alternatives.

### 6.5 Exploratory Low-resource Sarcasm

**OBSERVED EVIDENCE.** P2 compares PhoBERT fine-tune and ViPragSent (ours) at sarcasm-positive budgets of 64, 128, 256, 512, and 1,024, corresponding to total training sizes of 256, 512, 1,024, 2,048, and 4,096 under the verified 1:3 positive-to-other composition. Every system–budget condition has three recorded seeds; Appendix Table 6 reports the aggregate sarcasm macro F1 and normal 95% intervals. The PhoBERT means are 45.24, 45.15, 75.72, 75.41, and 76.34, whereas the ViPragSent means are 47.45, 47.79, 64.44, 74.35, and 74.89 across the increasing positive budgets. The PhoBERT sequence is non-monotonic, whereas the recorded ViPragSent sequence increases across the five budgets; the ViPragSent interval at 256 is wide [45.57, 83.30].

**PLAUSIBLE INTERPRETATION.** The

mixed patterns across systems show that the recorded positive-budget increase does not yield one uniform response across the two systems. They provide a three-seed exploratory characterization of these sampled training subsets, not a comparative data-efficiency conclusion.

**UNTESTED HYPOTHESIS.** The changes could depend on subset composition, optimization variation, or other unrecorded factors. The present design does not isolate those alternatives or establish an architectural explanation.

## 7 Conclusion

ViPragSent addresses two questions: how a frozen label-wise system compares with complete best-tuned baselines, and how aggregate performance should be read alongside label-specific leadership. On the fixed six-label evaluation, ViPragSent-Final reaches 84.3883 macro pragmatic F1, exceeding Vistral-7B by 1.5633 points. The result is an aggregate lead, not universal dominance: irony, idiom/figurative language, and code-switching remain narrowly below their respective leaders.

The contribution is a provenance-bounded evaluation resource, a documented development-only selection and freeze protocol, and traceable comparison artifacts rather than a claim of a universally superior architecture. The highest-priority next steps are controlled component ablations under the frozen protocol, independent reproduction when checkpoints are available, and resolution of source authorization and safe-access arrangements. Stored predictions, configuration, and hashes support inspection of the recorded evidence, but not an independent full training reproduction.

## 8 Limitations

The primary limitation is the data-access and provenance boundary. The fixed corpus combines 10,000 retained local social-media export records with 2,000 author-created, context-augmented derivatives based on VIVID idiom/proverb materials. Raw and processed source text remains private and is not redistributed; source authorization and license documentation are pending project action. The paper reports aggregate artifacts and provenance rather than public raw text, and it cannot offer an open-data claim or a public qualitative-example set. The AIVIVN artifact is a hash-identified mirror rather than a verified organizer-authored canonical release. These boundaries limit public data access and the availability of qualitative examples.

The main conclusions are also bounded by one adjudicated Vietnamese-language dataset with a documented social-media source component and one operational task formulation. The six pragmatic labels define the measured setting, not a complete account of Vietnamese pragmatics or a guarantee of behavior in another domain, population, label scheme, or deployment context. ViSoBERT is now an evaluated baseline, but this completed comparison does not make its result transferable beyond the recorded test set and protocol.

Several experimental qualifications further limit interpretation. The multi-task ablation is now three-seed but remains a configuration comparison rather than a causal or universal auxiliary-task effect. P1 is observational and uses unequal source strata, including only 334 VIVID-labelled records. P2 has three seeds per budget, mixed monotonicity across its two systems, and wide intervals in some cells, so it cannot establish data-efficiency superiority. Prompted API baselines are single recorded runs rather than multi-seed training results. Calibration is available only for systems with stored confidence scores. Finally, the study has no manual rationale-faithfulness verification and no permission-safe qualitative error examples. It therefore does not claim faithful rationales, broad retention or transfer, universal pragmatics coverage, or stable low-resource advantages.

ViPragSent-Final adds further limitations. It is a development-selected label-wise ensemble rather than a single model, and its thresholds and source assignments may be sensitive to the development protocol. The promoted configuration was evaluated once after freezing on one fixed Vietnamese test set. Its highest aggregate score does not imply strict per-label superiority: three localized deficits remain. Finally, the current verification is artifact-level; recorded checkpoint hashes are traceable, but the unavailable checkpoint files prevent an independent rerun.

## 9 Ethical Considerations

ViPragSent is handled as private, non-commercial research data. Its fixed gold corpus contains retained local ViSoBERT-export records and author-created, context-augmented derivatives based on VIVID idiom/proverb materials. Raw and processed source text are not redistributed, and source authorization and license documentation are pending project action. The manuscript reports aggregate results, task definitions, and provenance records without publishing raw text or source-text examples. External evaluation datasets are not components of ViPragSent; they are used only for the reported diagnostic and are not redistributed by this project.

The source-stratified P1 analysis does not change that governance boundary. The existence of aggregate results for a VIVID-labelled stratum is not evidence of VIVID license clearance, research-use authorization, unrestricted access, or a right to redistribute source or derivative text.

This boundary is intended to reduce privacy and contextual-integrity risks associated with sensitive user-generated language. Even when text is accessible to researchers, reproducing a post, attributing it to an individual, or exposing a rare linguistic pattern can increase identifiability or cause contextual harm. The project accordingly limits the paper to aggregate reporting and preserves data-access decisions outside the anonymous manuscript. Any future controlled-access or release mechanism should be assessed for governance and re-identification risk before it is implemented.

The annotation protocol uses two reviewers and a fixed adjudicator whose post-adjudication labels define the recorded gold splits. This process does not eliminate ambiguity in pragmatic interpretation, nor does it ensure that all social groups, dialects, or communicative norms are represented equally. Misclassification of irony, sarcasm, implicit sentiment, or other pragmatic behavior could misrepresent speakers or amplify unequal treatment when used in moderation, profiling, customer-service, or surveillance settings. The reported systems should therefore not be used as sole decision makers about individuals. The artifact-level traceability provided here supports inspection of recorded evidence, but it does not remove these representational risks or replace governance review in a deployment setting.

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# A Annotation Agreement

This appendix reports the per-label post-adjudication agreement values summarized in Section 3.

| Label | Records | Mean pairwise | Cohen's $\kappa$ | Fleiss' $\kappa$ | Krippendorff's $\alpha$ |
|---|---|---|---|---|---|
| implicit_sentiment | 12,000 | 0.95 | 0.80 | 0.79 | 0.79 |
| sarcasm | 12,000 | 0.94 | 0.79 | 0.79 | 0.79 |
| irony | 12,000 | 0.98 | 0.86 | 0.86 | 0.86 |
| idiom_figurative | 12,000 | 0.99 | 0.95 | 0.95 | 0.95 |
| code_switching | 12,000 | 0.96 | 0.79 | 0.78 | 0.78 |
| mocking | 12,000 | 0.94 | 0.80 | 0.79 | 0.79 |
| polarity | 12,000 | 0.88 | 0.81 | 0.80 | 0.80 |
| emotion | 12,000 | 0.83 | 0.78 | 0.78 | 0.78 |
| Macro | 12,000 | 0.93 | 0.82 | 0.82 | 0.82 |

Table 4: Recorded post-adjudication agreement by label family. The fixed adjudicator's labels define the experiment gold split; this is not independent third-rater agreement.

# B Label Prevalence

| Label | Train ($n = 8{,}000$) | Dev ($n = 2{,}000$) | Test ($n = 2{,}000$) |
|---|---|---|---|
| implicit_sentiment | 928 | 232 | 232 |
| sarcasm | 1,390 | 348 | 348 |
| irony | 738 | 185 | 184 |
| idiom_figurative | 733 | 183 | 184 |
| code_switching | 722 | 181 | 181 |
| mocking | 1,222 | 305 | 305 |

Table 5: Positive pragmatic-label counts in the fixed gold partitions. Labels are multi-label and may co-occur; counts therefore do not sum to the partition size.

# C Source-stratified and Low-resource Results

| System | ViSoBERT-local | VIVID-labelled |
|---|---|---|
| ViPragSent | 52.56 [52.38, 52.73] | 79.84 [79.71, 79.96] |
| PhoBERT fine-tune | 60.60 [58.63, 62.57] | 87.21 [85.48, 88.95] |
| ViSoBERT fine-tune | 62.08 [60.69, 63.47] | 93.04 [90.55, 95.54] |
| *P2: sarcasm macro F1 (three seeds)* | | |
| Budget | PhoBERT | ViPragSent |
| 64 | 45.24 [45.24, 45.24] | 47.45 [44.84, 50.06] |
| 128 | 45.15 [44.97, 45.32] | 47.79 [46.84, 48.75] |
| 256 | 75.72 [74.92, 76.53] | 64.44 [45.57, 83.30] |
| 512 | 75.41 [74.48, 76.34] | 74.35 [74.30, 74.41] |
| 1024 | 76.34 [75.74, 76.93] | 74.89 [74.07, 75.72] |

Table 6: Supplementary three-seed analyses (mean macro F1 [normal 95% interval]). The upper panel is a descriptive source-stratified sensitivity analysis on 1,666 ViSoBERT-local and 334 VIVID-labelled test records; it displays ViPragSent, PhoBERT, and ViSoBERT. The lower panel reports sarcasm macro F1 for the two systems at the five verified positive-label budgets. Each budget uses a 1:3 positive-to-other training composition, giving total training sizes of 256, 512, 1,024, 2,048, and 4,096. Neither panel identifies a causal source effect or a data-efficiency advantage.

## D External Diagnostics

| System | UIT-VSFC polarity | UIT-VSMEC emotion | AIVIVN-2019 polarity |
|---|---|---|---|
| ViPragSent without rationale | 41.67 [37.49, 45.86] | 32.40 [28.79, 36.01] | 57.81 [55.66, 59.96] |
| ViPragSent without emotion | 44.56 [36.51, 52.61] | 7.20 [4.31, 10.09] | 63.58 [56.66, 70.50] |
| ViPragSent without polarity | 29.55 [19.46, 39.64] | 37.27 [27.31, 47.23] | 44.10 [27.21, 60.98] |
| ViPragSent without uncertainty | 43.10 [38.32, 47.87] | 34.68 [30.13, 39.24] | 60.30 [58.05, 62.55] |
| ViSoBERT fine-tune | 26.74 [21.06, 32.41] | 6.69 [3.20, 10.18] | 31.96 [20.45, 43.47] |

Table 7: Three-seed external diagnostic macro F1 (mean [normal 95% interval]) for the four P0 ablations and ViSoBERT. UIT-VSFC and AIVIVN-2019 use polarity macro F1; UIT-VSMEC uses emotion macro F1. These datasets have distinct label spaces and are evaluation-only diagnostics, not a pooled transfer result.

## E Final-System Progression

| Historical system version | Macro pragmatic F1 | Status |
|---|---|---|
| ViPragSent original | 73.7469 | historical baseline |
| ViPragSent previous optimized | 82.0699 | historical development version |
| ViPragSent final incumbent | 84.0904 | archived incumbent |
| ViPragSent final continuation | 84.3883 | promoted final system |

Table 8: Historical ViPragSent progression. These rows document development and must not be interpreted as independent external baselines.